The AlSamadani Score: A Fail-Closed Governance and Confidence-Calibration Framework for Recursive Multi-Agent Clinical Decision Support Under Data Sparsity in the ICU and CVICU

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Key Points

Question Can a single bounded score tell when a recursive multi-agent large language model (LLM) consensus in the intensive care unit is too poorly grounded in fresh physiological data to be shown to clinicians?

Findings In this methods and simulation study, the proposed AlSamadani Score ($S_{\text{AlSamadani}} \in [0, 100]$) combines temporal data sparsity, collective epistemic drift, dissent collapse, and a sparsity-dependent confidence ceiling. On 2 author-constructed vignettes, it graded a dense-data encounter Tier I ($S_{\text{AlSamadani}} = 95.7$) and blocked the consensus for a sparse admission-triage encounter (Tier III, $S_{\text{AlSamadani}} = 38.1$), for which arterial lactate had the largest counterfactual value of information (+17.4 points). In 1000 synthetic encounters, the score discriminated correct from incorrect consensus with an area under the receiver operating characteristic curve (AUROC) of 0.691 (95% CI, 0.657–0.725), versus 0.642 for raw consensus confidence.

Meaning These findings justify validating the score on real data with a prespecified protocol before any use at the bedside. They are not evidence of clinical benefit.

Abstract

Importance Multi-agent LLM systems are being proposed for critical care. Iterative debate can make agents look more consistent and more confident without adding any evidence. At intensive care unit (ICU) admission, when laboratory data are sparse or stale, that pattern risks anchoring and automation bias.

Objective To specify and test the internal behavior of a fail-closed governance score that decides when a multi-agent consensus may be displayed, should be displayed only as differentials, or must be withheld.

Design, Setting, and Participants A methods study with deterministic vignettes and a seeded Monte Carlo simulation ($N = 1000$ encounters across 4 sparsity and confidence-laundering scenarios), plus a scaling benchmark ($M \in \{3, 5, 7, 10\}$ agents, $R \in \{1, 2, 4, 6\}$ rounds) with and without a "catfish" dissenting agent. No patient data were used. Open-source code includes a loader for MIMIC-IV v3.1 and eICU-CRD for future validation.

Main Outcomes and Measures Governance tier, AUROC of $S_{\text{AlSamadani}}$ for correct consensus, expected calibration error (ECE) before and after the confidence ceiling, and dissent-collapse ratio.

Results The dense-data vignette was graded Tier I ($S_{\text{AlSamadani}} = 95.7$) and the sparse admission vignette Tier III ($S_{\text{AlSamadani}} = 38.1$; consensus withheld; deterministic sepsis-bundle fallback issued). In simulation, the AUROC was 0.691 (95% CI, 0.657–0.725) versus 0.642 for raw confidence. Consensus accuracy was 76.6% in Tier I and 31.9% in Tier III (4-class problem; chance, 25%). Capping confidence lowered ECE from 0.143 to 0.120. The catfish agent reduced dissent collapse and raised $S_{\text{AlSamadani}}$ but did not improve consensus accuracy. The dissent component can therefore be inflated without any epistemic gain.

Conclusions and Relevance The AlSamadani Score is a transparent, fail-closed rule for withholding under-grounded LLM consensus. Its weights are expert priors, and its performance figures are synthetic. Prospective, TRIPOD+AI–guided validation is required before clinical use.

Introduction

LLM agents are increasingly evaluated on ICU data. AgentRx benchmarked LLM agents on multimodal clinical risk prediction using MIMIC ICU data. In that study, single-agent configurations outperformed naive multi-agent baselines, and both trailed supervised fusion models.[1] Agentic systems now run complete emergency department workflows on MIMIC-IV cases.[2] Debate among agents has well-known benefits for factuality on benchmark tasks.[3] In medical question answering, however, Wang and Yang showed that debate can produce a *consistency illusion*. Agents stop contradicting each other while the similarity of their reasoning chains falls, as measured by their cross-agent reasoning alignment (CARA) metrics.[4] In a 3-agent retrieval-augmented system for early sepsis management evaluated by 2 intensivists, hallucinated clinical findings appeared in 3 of 10 cases.[5]

These observations matter most at ICU admission. At that point lactate, arterial blood gases, and coagulation studies are often pending or hours old. A confident consensus formed then may rest on population priors rather than patient data. Known failure modes of clinical decision support compound the risk. The most relevant are automation bias[6] and inflated or poorly calibrated outputs from widely deployed sepsis models.[7] Calibration metrics such as ECE[8] assess a model's confidence but say nothing about whether the evidence for that confidence is present at the moment it is displayed.

We propose the AlSamadani Score ($S_{\text{AlSamadani}}$), a bounded, fully specified governance layer that sits *between* a multi-agent system and the clinician. It answers 1 operational question: whether the consensus should be shown, shown only as differentials, or withheld. It also says which data to obtain. The score is *fail-closed*: any missing input, invalid trace, or software error resolves to abstention. Abstention is never empty: a deterministic guideline engine evaluated on observed data supplies a bedside fallback.

Methods

Setting and notation

An encounter e is evaluated at time t (hours since ICU admission). M specialist agents deliberate for R rounds over K candidate diagnoses. Agent m in round r emits a probability vector $p_{m,r} \in \Delta^{K-1}$; round 0 holds the independent first opinions. The consensus is $\bar{p}_r = \frac{1}{M}\sum_m p_{m,r}$ and its confidence is $c_r = \max_k \bar{p}_{r,k}$. All divergences use the base-2 Jensen–Shannon divergence $\text{JSD}_2 \in [0, 1]$.[9]

Table 1: Biomarker panel, prespecified weights, and half-lives

j	Biomarker (unit)	w_j	h_j, h	MIMIC-IV itemid[a]
1	Arterial lactate (mmol/L)	0.22	2	50813
2	PaO_2/FiO_2 ratio (mm Hg)	0.15	4	50821 + chart 223835
3	Serum creatinine (mg/dL)	0.13	12	50912
4	Platelet count ($\times 10^9$/L)	0.11	24	51265
5	White blood cell count ($\times 10^9$/L)	0.10	12	51301
6	International normalized ratio	0.10	24	51237
7	Bicarbonate (mmol/L)	0.10	6	50882
8	Total bilirubin (mg/dL)	0.09	24	50885

[a]Default identifiers in the loader. Verify them against the local `d_labitems`/`d_items` dictionaries; eICU-CRD lab names are listed in the code.

Temporal-Physiological Sparsity Index

Eight canonical sepsis and Sequential Organ Failure Assessment (SOFA) biomarkers[10, 11] carry decision weights w_j ($\sum_j w_j = 1$) and decision-relevance half-lives h_j (Table 1). If $m_j \in \{0, 1\}$ indicates that biomarker j has been observed and Δt_j is its elapsed age in hours, then

$$\sigma(x_t) = 1 - \sum_{j=1}^{8} w_j \, m_j \, e^{-\lambda_j \Delta t_j}, \qquad \lambda_j = \frac{\ln 2}{h_j}, \qquad \sigma \in [0, 1]. \tag{1}$$

The index is 0 when every biomarker is observed at $\Delta t_j = 0$ and 1 when none is available. The half-lives encode how fast each value loses decision relevance in shock, not its pharmacokinetics. The weights and half-lives are expert priors set by the author before any simulation. They were not fitted to data.

Collective Epistemic Drift

Drift is the cumulative change in the consensus across rounds plus a penalty for *confidence laundering*, ie, confidence that rises faster than grounded evidence:

$$\mathrm{CED}_R = \min\Big\{1, \sum_{r=1}^{R} \mathrm{JSD}_2(\bar{p}_{r-1}, \bar{p}_r) \; + \; \gamma \, \max\big(0, \; \Delta c - \Delta I_{\mathrm{evid}}\big)\Big\}, \tag{2}$$

where $\Delta c = c_R - c_0$ and $\Delta I_{\mathrm{evid}} = I_R - I_0$, with I_r the grounded-evidence information available at round r (eg, $1 - \sigma$ if new results arrive during deliberation; otherwise constant, so $\Delta I_{\mathrm{evid}} = 0$). We set $\gamma = 1$.

Dissent Collapse Ratio under sparsity

With mean pairwise disagreement $D_r = \binom{M}{2}^{-1} \sum_{m<m'} \mathrm{JSD}_2(p_{m,r}, p_{m',r})$,

$$\rho_{\mathrm{collapse}} = \max\big(0, \, 1 - D_R / D_0\big) \quad (\rho = 0 \text{ if } D_0 < 10^{-3}). \tag{3}$$

Agents converging on dense data is expected and benign. Convergence on sparse data is the warning sign. The score therefore uses the sparsity-weighted product $\sigma\rho$.

Sparsity-Penalized Calibration Error and the Epistemic Confidence Ceiling

The confidence the available data can support is capped at

$$c_{\max}(\sigma) = 1 - \kappa\sigma, \qquad \mathrm{SPCE}_e = \frac{\max\big(0, \, c_R - c_{\max}\big)}{\kappa} \in [0, 1], \tag{4}$$

Table 2: Governance tiers and bedside actions

Tier	$S_{\text{AlSamadani}}$	Display policy
I, Grounded	≥ 80.0	Consensus diagnosis with ceiling-capped probability $\min(c_R, c_{\max})$; standard oversight; top-3 VOI shown.
II, Guarded	50.0–79.9	No single-answer framing. Top-2 differentials, round-0 dissenting rationales, and top-3 VOI laboratory refresh recommendations.
III, Mandatory AI abstention	< 50.0 or any failure	`consensus_diagnosis_index = None`; no probability shown. Deterministic SSC 2026/KDIGO fallback on observed data plus stat laboratory tests ranked by VOI.

with $\kappa = 0.5$. Thus a fully sparse encounter cannot display a probability above 0.5. At cohort level, $\text{SPCE} = \text{ECE}\big(\min(c_i, c_{\max,i})\big) + \overline{\text{SPCE}_e}$, combining the residual miscalibration of capped confidences[8, 12] with the mean excess over the ceiling.

The AlSamadani Score

$$S_{\text{AlSamadani}} = 100\Big[1 - \big(\alpha_\sigma\,\sigma + \alpha_D\,\text{CED}_R + \alpha_\rho\,\sigma\rho_{\text{collapse}} + \alpha_C\,\text{SPCE}_e\big)\Big], \tag{5}$$

with prespecified weights $(\alpha_\sigma, \alpha_D, \alpha_\rho, \alpha_C) = (0.40, 0.20, 0.20, 0.20)$. *Boundedness.* Each component lies in $[0, 1]$ and the α weights sum to 1, so $S_{\text{AlSamadani}} \in [0, 100]$. *Monotonicity.* For a fixed deliberation trace, $\partial S_{\text{AlSamadani}}/\partial\sigma = -100\big(\alpha_\sigma + \alpha_\rho\rho + \alpha_C \mathbb{1}[c_R > c_{\max}]\big) \leq 0$ almost everywhere. Fresher or more complete data never lower the score.

Counterfactual value of information

For each biomarker, $\text{VOI}_j = S_{\text{AlSamadani}}(x_t \mid \Delta t_j{=}0,\, m_j{=}1) - S_{\text{AlSamadani}}(x_t)$. The deliberation trace is held fixed, so VOI measures how much of the score the governance layer would recover if that test were drawn now; it does not predict how the agents would update. By Eq. 5, while the ceiling stays active, $\text{VOI}_j = 100\,w_j(1 - m_j e^{-\lambda_j \Delta t_j})(\alpha_\sigma + \alpha_\rho\rho + \alpha_C)$. VOI therefore scales with each biomarker's lost information multiplied by the encounter's collapse level. Tests are ranked by VOI_j.

Governance tiers and the fail-closed fallback

Table 2 shows the tiers. In Tier III the software sets `consensus_diagnosis_index = None` and returns no probability. The aim is to prevent anchoring and to spare clinicians the work of verifying an under-grounded answer. Validation errors, missing traces, or exceptions also resolve to Tier III. In every tier a deterministic rule engine evaluates the *observed* vital signs and laboratory values against the 2026 Surviving Sepsis Campaign (SSC) adult guideline[13] and the KDIGO acute kidney injury guideline.[14] The SSC rules cover antimicrobials ideally within 1 hour when shock is possible, lactate measurement when it is unavailable, and individualized balanced crystalloid resuscitation (30 mL/kg considered within 3 hours, reassessed with dynamic measures). They also cover norepinephrine as first-line vasopressor with an initial mean arterial pressure (MAP) target of 65 mm Hg (60–65 mm Hg at age 65 years or older) and lung-protective ventilation at 6 mL/kg predicted body weight. The KDIGO rules cover AKI staging with nephrotoxin avoidance. The engine never imputes values. It returns at least 1 recommendation by construction, which we call guideline coverage; coverage is not evidence of efficacy. Every recommendation is a prompt for clinician review, not an order.

Box. Runtime governance pipeline (one evaluation)

1. *De-identify*: replace patient, stay, and unit identifiers with HMAC tokens; convert timestamps to Δt_j; scrub free text; abort (Tier III) if a residual-PHI check fails or no salt is configured.
2. *Ground*: compute $\sigma(x_t)$ from the observed biomarker ages (Eq. 1).
3. *Deliberate* (external multi-agent system): collect $p_{m,r}$ for $r = 0, \ldots, R$ and the round-0 rationales.
4. *Audit*: compute CED_R, ρ_{collapse}, $c_{\max}$, and SPCE_e (Eqs 2–4), then $S_{\mathrm{AlSamadani}}$ (Eq. 5).
5. *Decide*: assign the tier (Table 2); on any exception, assign Tier III and set `consensus_diagnosis_index = None`.
6. *Act*: always run the deterministic SSC 2026/KDIGO engine on observed data; rank tests by VOI_j.
7. *Log*: store the tokens, the component values, the tier, and the fallback rule identifiers for audit. Store no raw identifiers.

Privacy architecture

Patient and unit identifiers are replaced by salted, domain-separated HMAC-SHA256 tokens of the form `ANON-XXXXXXXXXXXXXXXX` (16 hexadecimal characters). The salt is held outside the model context. If no salt is configured, the software refuses to run (fail-closed). A regular-expression scrubber redacts names, record numbers, Saudi national ID and Iqama numbers, telephone numbers, email addresses, calendar dates, and facility names. All times are converted to elapsed hours (Δt_j) relative to ICU admission. A final guard raises an error if any personal health information pattern or raw identifier survives into the prompt. These are technical safeguards only. Lawful use requires local governance under the Saudi Personal Data Protection Law (PDPL), policies of the Saudi Data and Artificial Intelligence Authority (SDAIA), and, for MIMIC-IV and eICU-CRD, PhysioNet credentialing and the data use agreement. The agreement restricts transmitting data to third-party online services.[15, 16]

Simulation design

The simulation was prespecified, and all parameters were fixed before the first run (seed 20261110). In each encounter, the true diagnosis y was drawn from $K = 4$ classes. Each biomarker was observed with probability $q \in \{0.95, 0.60, 0.25, 0.25\}$ (scenarios S0–S3), with age drawn from $\mathcal{U}(0, kh_j)$, $k \in \{1, 2, 3, 3\}$. Agent logits were $\beta e_y + b + \varepsilon_m$, where the signal was $\beta = 3(1 - \sigma)$, $b \sim \mathcal{N}(0, I)$ was a bias shared by all agents (common model family), and $\varepsilon_m \sim \mathcal{N}(0, I)$ was agent-specific noise. Each round applied conformity and sharpening, $p \leftarrow \mathrm{norm}\big[((1 - \eta)p + \eta\bar{p})^{1/\tau}\big]$, with $\eta = 0.5$. We used $\tau = 0.9$ in S0–S2 and $\tau = 0.6$ in S3 to model confidence laundering. The base case had $M = 5$ and $R = 2$. The outcome was a correct argmax consensus. We report the AUROC from the Mann-Whitney statistic with a 1000-replicate bootstrap 95% CI. In the scaling benchmark, run under S3 conditions with 200 encounters per cell and common random numbers across arms, the catfish agent kept its independent round-0 distribution in every round.

Results

Vignettes

Case A had complete laboratory data, each value 1 hour old or less, and its agents agreed early. Case B was an admission triage at $T = 0$ hours with no lactate, blood gas, coagulation, or liver tests, and with 3 laboratory values 3 hours old. Its agents disagreed initially, then converged with rising confidence and no new evidence (Table 3). Case A scored $S_{\mathrm{AlSamadani}} = 95.7$ (Tier I). Case B scored $S_{\mathrm{AlSamadani}} = 38.1$ (Tier III). For Case B the consensus was withheld, and the fallback issued the antimicrobial, stat-lactate, fluid, MAP, and ventilation prompts. Case B's laundering term was 0.510: its confidence of 0.840 exceeded its ceiling of 0.647. Arterial lactate

Table 3: Component values for the 2 vignettes and counterfactual VOI ranking for Case B

Case	σ	CED_R	ρ	c_R	$c_{\max}$	$S_{\text{AlSamadani}}$	Tier	Consensus shown
A, dense	0.058	0.050	0.897	0.804	0.971	95.7	I	Yes
B, sparse triage	0.706	0.629	0.950	0.840	0.647	38.1	III	No (`None`)

Rank	Biomarker refreshed (Case B)	VOI_j, points	$S_{\text{AlSamadani}}$ if refreshed
1	Arterial lactate	+17.4	55.5
2	PaO2/FiO2 ratio	+11.9	49.9
3	Bicarbonate	+7.9	46.0
4	INR	+7.9	46.0
5	Total bilirubin	+7.1	45.2
6	Serum creatinine	+1.6	39.7
7	White blood cell count	+1.3	39.3
8	Platelet count	+0.7	38.8

Table 4: Synthetic Monte Carlo results by scenario ($M = 5$, $R = 2$)

	Scenario	n	Mean σ	Mean $S_{\text{AlSamadani}}$	Acc., %	I, %	II, %	III, %
S0	dense, fresh labs; mild sharpening	250	0.323	79.3	76.4	49.6	50.0	0.4
S1	moderate sparsity	250	0.666	58.6	52.4	0.0	87.6	12.4
S2	high sparsity, no laundering	250	0.896	44.4	30.4	0.0	19.6	80.4
S3	high sparsity + confidence laundering	250	0.894	38.9	34.8	0.0	18.8	81.2

ranked first by VOI (+17.4 points), followed by the PaO2/FiO2 ratio (+11.9) and bicarbonate (+7.9). These values match the closed form in Methods: $0.22 \times (0.40 + 0.20 \times 0.950 + 0.20) \times 100$ for lactate.

Monte Carlo simulation

Across 1000 synthetic encounters, $S_{\text{AlSamadani}}$ discriminated correct from incorrect consensus with an AUROC of 0.691 (95% CI, 0.657–0.725). Raw consensus confidence had an AUROC of 0.642. Mean scores and tier distributions followed the scenario gradient (Table 4). Consensus accuracy was 76.6% in Tier I ($n = 124$) and 31.9% in Tier III ($n = 436$); $n = 440$ encounters fell in Tier II. Capping confidence at $c_{\max}$ reduced ECE from 0.143 to 0.120.

Scaling and the catfish agent

Without a dissenter, ρ_{collapse} rose toward 1 as rounds increased at every panel size, and $S_{\text{AlSamadani}}$ fell accordingly (Table 5). A single catfish agent kept ρ substantially lower and raised mean $S_{\text{AlSamadani}}$ by up to about 25 points. Consensus accuracy, however, was essentially unchanged (differences of 1.0 percentage point or less in every cell). In this generative model, preserved dissent therefore restored the *score* without improving the *answer*.

Table 5: Scaling benchmark under scenario S3 (200 encounters per cell)

M	R	ρ naive	ρ catfish	$S_{\text{AlSamadani}}$ naive	$S_{\text{AlSamadani}}$ catfish	Acc. naive, %	Acc. catfish, %
3	1	0.531	0.294	48.2	54.3	36.0	36.0
3	2	0.778	0.298	37.6	50.9	34.0	33.0
3	4	0.956	0.261	25.5	47.2	34.5	35.5
3	6	0.992	0.235	19.6	44.5	33.5	34.5
5	1	0.496	0.360	48.9	52.5	40.5	40.0
5	2	0.748	0.457	38.9	46.8	33.5	33.0
5	4	0.946	0.457	26.0	40.5	35.5	36.0
5	6	0.992	0.364	19.2	37.7	33.0	32.5
7	1	0.490	0.382	49.2	51.9	37.0	37.0
7	2	0.747	0.540	38.8	44.6	35.0	34.0
7	4	0.947	0.528	25.9	38.1	30.0	30.5
7	6	0.988	0.504	19.9	34.0	37.5	38.0
10	1	0.479	0.413	50.9	52.6	40.0	40.0
10	2	0.744	0.604	39.7	43.6	33.5	33.5
10	4	0.947	0.654	25.8	34.4	31.5	31.5
10	6	0.989	0.620	19.7	30.1	28.5	28.0

Discussion

The AlSamadani Score turns 4 failure signatures of recursive multi-agent deliberation into a single, auditable display rule: missing or stale data, drift unsupported by evidence, collapse of dissent under sparsity, and confidence above what the data support. Its distinguishing design choice is that abstention is *active*. When the AI is silenced, the clinician receives the guideline bundle and a ranked list of the tests most likely to restore a usable assessment. In the sparse admission vignette, arterial lactate topped that list. Lactate carries the largest weight and the shortest half-life in the panel, and SSC 2026 places it at the center of early resuscitation.[13]

The simulation supports the internal logic of the score. Scores were lower when data were sparser, and the tiers separated consensus accuracy. The confidence ceiling also reduced calibration error. The AUROC of 0.691 is modest, however, and it is a property of our generative model, not of any real agent system. We report it as such, without tuning the parameters to reach a target value.

The most important finding is negative. A stubborn dissenter lowers ρ_{collapse} by construction and so raises $S_{\text{AlSamadani}}$ without improving accuracy. A naive implementation could therefore be gamed: adding a contrarian agent would move encounters out of Tier III with no epistemic gain. This mirrors the consistency illusion described by Wang and Yang[4] in the opposite direction, where *apparent* diversity is mistaken for grounded diversity. Before validation, we recommend 3 changes: (1) require catfish dissent to cite observed data, for example by weighting D_r with a reasoning-alignment measure such as CARA;[4] (2) exclude designated dissenters from D_r; or (3) cap the credit the dissent term can contribute.

Limitations

First, no patient data were analyzed. All performance figures are synthetic and depend on the assumed generative model. Second, the weights, half-lives, γ, κ, and the tier thresholds are expert priors and have not been calibrated. Third, VOI is first order: it holds the agents' beliefs

fixed. Fourth, the panel omits the nonlaboratory SOFA components and cardiac-specific markers that matter in the cardiovascular ICU (CVICU), such as troponin, natriuretic peptides, and mixed venous saturation. Fifth, the regular-expression scrubber is not a certified de-identification tool. Sixth, the guideline engine paraphrases SSC 2026 and KDIGO, and its wording and recommendation strengths must be checked against the primary texts before deployment.

Planned validation

We will register a protocol for retrospective validation. It will use MIMIC-IV v3.1 for development and eICU-CRD for external evaluation, with the bundled loader. We will report it according to TRIPOD+AI[17] (eTable). Prespecified outcomes are discrimination, calibration, and net benefit of $S_{\text{AlSamadani}}$ for consensus error, together with the time to and yield of VOI-ranked tests. Prospective silent-mode evaluation at the authors' institutions would follow, subject to ethics approval.

Regulatory considerations

We designed the software with reference to the 4 nondevice clinical decision support criteria of FD&C Act §520(o)(1)(E), as added by 21st Century Cures Act §3060(a), and the US Food and Drug Administration's 2026 clinical decision support guidance.[18] Time-critical ICU use can place software outside those criteria. We make no regulatory determination; formal review is required. The design also draws on World Health Organization guidance on AI for health[19] and on Saudi PDPL and SDAIA requirements.

Conclusions

The AlSamadani Score is a transparent, fail-closed rule for withholding under-grounded multi-agent LLM consensus in the ICU. It always provides a guideline fallback and data-acquisition guidance in its place. The score is ready for prespecified validation. It is not ready for clinical use.

Article Information

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Use of AI (JAMA Network, ICMJE, arXiv/alphaXiv, and Saudi AI Principles in Media): Generative AI tools assisted with this work. Google Gemini was used for preliminary conceptual exploration. Claude (Anthropic), through Claude Code, was used in November 2026 to draft and refactor the software, the unit tests, and this manuscript text, and to search for and screen literature. All equations, parameter choices, and claims were reviewed by the author, who accepts full responsibility for the content, consistent with JAMA Network guidance,[20] ICMJE recommendations,[21] the arXiv policy on generative AI,[22] and the disclosure and human-responsibility principles of the Saudi *AI Principles in Media*.[23] AI tools are not authors. No patient data were entered into any AI tool. [The author must confirm this statement after reviewing the final text.]

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eTable. Reporting status against TRIPOD+AI[17] domains for this methods and simulation study

TRIPOD+AI domain	Status	Where addressed or plan
Title and abstract	Reported	Title; structured abstract; Key Points
Background and objectives	Reported	Introduction
Data sources and participants	Not applicable	Synthetic data only; MIMIC-IV v3.1 and eICU-CRD planned
Outcome definition	Partly reported	Simulated correct consensus; clinical reference standard to be prespecified
Predictors and their handling	Reported	Table 1; Eq. 1; loader specification
Sample size	Not applicable	Simulation $N = 1000$; formal sample-size calculation planned for validation
Missing data	Reported (design)	Missingness is a model input (m_j) and is never imputed
Model specification	Reported	Eqs 1–5; prespecified parameters; open code
Performance (discrimination, calibration)	Synthetic only	Table 4; real-data evaluation pending
Fairness and subgroups	Pending	Planned across age, sex, unit type (ICU vs CVICU), and site
Model updating and monitoring	Pending	Silent-mode prospective evaluation
Open science and AI use	Reported	Article Information; Data Sharing Statement
Limitations	Reported	Discussion

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